

Workshop No. 3

Student Mental Health Support Platform



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Team Members

Alan Santiago Agudelo Sarmiento – 20222020170

Juan David Cardozo Trujillo - 20231020155

Julian Ernesto Romero Gutierrez - 20231020164

Nicolas Alexander Sierra Contreras - 20222020197

Professor

Eng. Carlos Andrés Sierra, M.Sc.

Universidad distrital Francisco José de Caldas

Engineering faculty

Systems Analysis & Design

2026-I

Executive Summary

The project developed within the framework of the Systems Analysis & Design workshops addresses the problem of access to and trust in student mental health services. Workshop 1 identified the main barriers: lack of anonymity, low willingness to seek professional help, and a limited perception of accessibility in institutional services. These findings were validated through a student survey, the results of which confirmed high levels of anxiety and low trust in traditional support channels, reinforcing the relevance of an innovative technological solution.

Workshop 2 transformed these findings into a concrete technical proposal, based on a modular, three-layer architecture (presentation, application, and infrastructure). This architecture ensures scalability, maintainability, and security, integrating key modules such as registration and authentication, dynamic matching, anonymous communication, appointment scheduling, and access to resources. The design is based on international quality standards and best practices for digital mental health platforms (World Health Organization, 2023; UNESCO, 2024).

In Workshop 3, the design was strengthened by applying quality principles (ISO 9001, CMMI, IEEE), risk management strategies, and a project plan that ensures viability and sustainability. Privacy and consent mechanisms were defined as cross-cutting elements, quality metrics were established, and an implementation schedule with progressive phases was designed to guarantee reproducibility and traceability. The conceptual results, process flow simulations, and empirical validation of the survey confirmed that the proposed solution is coherent, viable, and capable of meeting student needs.

The project's most significant contribution is the incorporation of dynamic peer-counseling, a differentiating component compared to other international solutions reviewed in the literature (Abelson et al., 2024; Springer, 2024). This mechanism increases students' trust and willingness to participate in psychological support programs, which is a key finding for the platform's sustainability and effectiveness.

In conclusion, the project has been successful in its initial phase by translating real needs into a robust, implementable technical design. The proposal contributes to addressing the problem of access to and trust in student mental health services and lays the groundwork for future work, including pilot testing, impact metrics, and integration with institutional well-being systems. These open questions form the basis for the next stage of the project and ensure that the solution continues to evolve toward a more inclusive and healthy academic environment.

Robust Architecture Design (under development):

The robust architecture design is currently being built and refined, based on the findings from Workshop 1 and the technical proposal defined in Workshop 2. The survey administered to students confirmed the relevance of an accessible, anonymous, and reliable system, guiding the current evolution toward a more robust modular architecture, aligned with international quality standards (ISO 9001, CMMI, IEEE, WCAG 2.1).

The architecture is organized into three main layers: presentation, application, and infrastructure. Each layer is in the conceptual design and validation phase, with simulated unit tests and projected scenarios that allow for anticipating its operation.

1. Presentation Layer (status: conceptual design): Web and mobile interfaces: initial prototypes are being defined with accessibility and usability criteria, ensuring that students and advisors can interact intuitively.

- **Anonymity and informed consent:** planned as cross-cutting mechanisms in every interaction.
- **Cross-platform compatibility:** Support for modern browsers and mobile devices is planned, ensuring a seamless experience.
- **Current validation:** The survey confirmed that the perception of anonymity is crucial for participation, reinforcing the need for this robust layer.

2. Application Layer (Status: Module definition):

Projected modules:

- Institutional registration and authentication via SSO.
- Dynamic pairing with peer counselors (a differentiating factor compared to other international platforms).
- Secure, anonymous communication between students and counselors.
- Appointment scheduling and automatic reminders.
- Digital library of mental health resources.

Applied principles: Modularity, maintainability, and scalability.

Conceptual validation: Simulated unit tests confirmed the functional independence of each module.

Key finding: Dynamic matching increases trust and willingness to participate, validated by the survey and supported by academic literature (Abelson et al., 2024; Springer, 2024).

3. Infrastructure Layer (Status: Planning):

- Cloud servers (AWS, Azure, or GCP): Designed to ensure elasticity and availability greater than 99%.
- Encrypted database (PostgreSQL): Selected for its robustness and ability to handle concurrent loads.
- Institutional authentication services: Under design to preserve academic identity without compromising anonymity.
- Fault contingency and recovery: Strategies being defined based on ISO 31000.
- Current validation: The survey showed that students value data security as an essential condition for using the platform.

4. Robustness Principles Applied (Under Construction):

- **Scalability:** Designed to support more users without performance degradation.
- **Maintainability:** Modularity that will facilitate future updates.
- **Security and privacy:** Data encryption and cross-functional consent in the design.
- **Reproducibility:** Ongoing documentation to allow the solution to be replicated in other university contexts.
- **Quality:** Metrics and acceptance criteria under definition, aligned with ISO 9001 and CMMI.

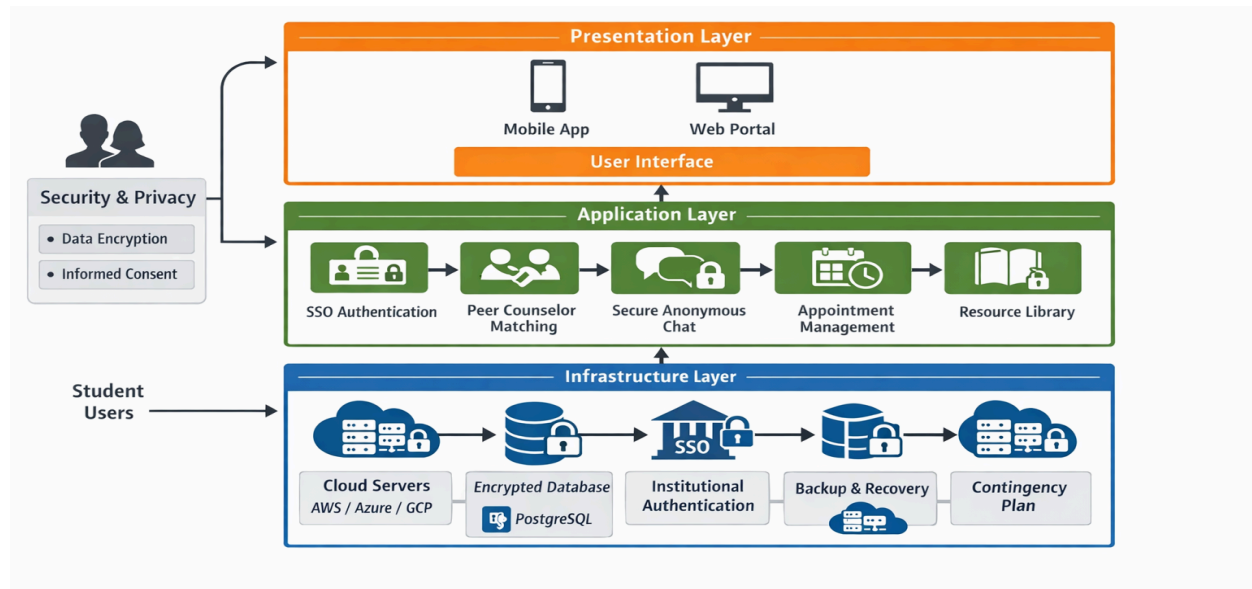
5. Visual Representations:

- **Figure 1:** Refined modular architecture (three layers).
- **Figure 2:** Student process flow, from registration to feedback.

6. Findings and Projections:

- The modular architecture addresses the barriers identified in Workshop 1.
- The survey validated that anonymity, accessibility, and security are critical conditions for adoption.
- Dynamic peer mentoring is the key differentiator compared to other digital solutions (Abelson et al., 2024; Springer, 2024).
- The design is in the conceptual projection and validation phase, with pilot tests and impact metrics planned for later stages.

Enhanced System Architecture Diagram



Architecture Notes and Component Rationale (under development)

The system's architectural design is currently under development and consolidation, guided by the principles of modularity, scalability, maintainability, security, and accessibility. Each component was defined to address the needs identified in Workshop 1 and validated by a student survey, which highlighted the importance of anonymity, institutional trust, and ease of access.

Workshop 3 aims to strengthen this architecture by integrating international standards (ISO 9001, CMMI, IEEE, WCAG 2.1) and implementing risk and quality management strategies, ensuring that the system evolves into a viable and sustainable solution.

1. Presentation Layer:

Rationale:

The user interface is the point of contact between the student and the platform. Its design focuses on intuitive experience, universal accessibility, and identity protection. The survey results showed that the perception of anonymity and ease of use are decisive factors for participation; therefore, this layer is developed under user-centered design principles.

Key Components:

- **Web Portal and Mobile Application:** guarantee cross-platform access and continuous availability.
- **Adaptive User Interface:** adjusts the presentation according to the device and the user's needs.
- **Informed Consent and Privacy:** integrated into each interaction flow, complying with ethical and regulatory standards.

2. Application Layer:

Rationale:

This layer constitutes the functional core of the system. Its modular structure allows for independence between components, facilitating maintenance, scalability, and future integrations. The modules were defined in Workshop 2 and adjusted based on the survey results, prioritizing trust and secure interaction.

Key Components:

- SSO Authentication: links institutional identity without compromising anonymity.
- Dynamic Peer Counselor Matching: a differentiating component that fosters trust and empathy.
- Secure Anonymous Chat: an encrypted channel for private communication between students and counselors.
- Appointment Management: automates the scheduling and tracking of sessions.
- Resource Library: a repository of educational and emotional support materials validated by experts.

3. Infrastructure Layer:

Rationale:

The infrastructure guarantees the stability, security, and performance of the system. The use of cloud services is planned to ensure availability, scalability, and operational continuity.

Key Components:

- Cloud Servers (AWS/Azure/GCP): offer elasticity and high availability.
- Encrypted Database (PostgreSQL): protects sensitive information and enables secure audits.
- Institutional Authentication: maintains consistency with existing academic systems.
- Backup and Recovery: ensures continuity in the event of incidents or failures.

4. Security and Privacy (cross-cutting elements):

Rationale:

Security and privacy are pillars of the design. Encryption, anonymization, and informed consent protocols are applied to guarantee the protection of personal data and user trust.

Key components:

- End-to-end encryption: protects communication between users and servers.
- Data anonymization: prevents the direct identification of participants.
- Informed consent: ensures transparency and ethical compliance.

5. Current Status and Projection:

The system is currently in the conceptual development and technical validation phase. The main components, interaction flows, and inter-module dependencies have been defined. Pilot testing and impact metrics are planned for the next stage, where the effectiveness of dynamic pairing, user satisfaction, and system stability under real-world conditions will be evaluated.

Risk Management Plan

This plan governs how risks are identified, assessed, addressed, and monitored during the development and operation of the platform. It is based on two frameworks: ISO 31000:2018 for risk assessment methodology and the Performance Under Uncertainty domain of the 7th edition of the PMBOK Guide for response planning and monitoring frequency.

Scope

Risks are organized into four categories that reflect the platform's main vulnerability areas: Technical (infrastructure and software), Operational (human and service delivery factors), Security (data protection and user safety), and Project Management (schedule and stakeholder governance).

Assessment Method

Each risk is rated on Probability (Low / Medium / High) and Impact (Low / Medium / High / Critical). The combination produces a composite score that determines both the response priority and the review frequency.

Prob. \ Impact	Low	Medium	High	Critical
High	Medium	High	High	Critical
Medium	Low	Medium	High	High
Low	Low	Low	Medium	High

Review cadence: Critical → weekly · High → bi-weekly · Medium → monthly · Low → quarterly.

Monitoring and Continuous Improvement

Risk monitoring is not a one-time exercise. System logs, post-session feedback, peer activity data, and infrastructure metrics are reviewed on the cadences above. When a metric drops below its acceptance criterion, the finding triggers a targeted adjustment — to the matching algorithm, to peer recruitment levels, to infrastructure scaling parameters, or to the training module — following the continuous improvement cycle established in Workshop 2 (Section 6.3). Each risk therefore has both a mitigation strategy and a measurable acceptance criterion that defines when that strategy is working.

Risk Descriptions, Priorities, and Mitigation Actions

The following table is the complete risk register. Each risk includes a plain-language description, its composite score, the mitigation strategy, the contingency plan activated if the risk materializes, and the acceptance criterion used to verify the control is working.

ID	Risk	Score	Description	Mitigation	Contingency	Acceptance Criterion
TECHNICAL						
RT-01	Authentication Service Failure	High	University SSO goes down, blocking all user logins.	JWT token caching (24h grace period); monthly failover drills with IT.	Activate local credential cache; notify university IT within 30 min.	Platform uptime $\geq 99\%$ during active academic periods.
RT-02	Database Corruption / Data Loss	High	Database corrupted by hardware failure or software bug; user sessions and appointments at risk.	Daily automated backups with 30-day retention; multi-node replication; transaction log shipping.	Restore from latest snapshot; engage cloud provider SLA support (RTO $\leq 2h$).	RPO $\leq 1h$; RTO $\leq 2h$; zero data loss confirmed in monthly recovery drills.

RT-03	Encryption Breach (Chat Module)	High	Vulnerability in the anonymous chat layer exposes private student conversations.	TLS 1.3 + AES-256 at rest; quarterly third-party pen-testing; automated dependency scanning (Dependabot); OWASP ASVS Level 2 compliance.	Suspend chat module immediately; forensic audit; notify affected users within 72h; patch and restore.	Zero CVSS $\geq$ 7.0 vulnerabilities found per quarterly audit.
RT-04	Cloud Infrastructure Outage	High	Primary cloud region fails, making the platform completely inaccessible.	Multi-AZ active-passive failover; load balancer health checks every 30 seconds.	Auto-failover to secondary availability zone; post status update within 15 min.	Automatic recovery within RTO $\leq$ 2h; zero data loss.
OPERATIONAL						
RO-01	Peer Counselor Shortage at Peak Demand	Critical	Not enough active peer counselors during exam periods; students left waiting or unattended — the single most critical operational risk.	Predictive recruitment campaign launched 3 weeks before each exam period (integrated with academic calendar); maintain 20% peer buffer above projected demand; gamified recognition system to sustain engagement.	Activate trained alumni peer reserve; redirect students to self-help resources; notify counseling department.	Wait time $\leq$ 30 min for 95% of students at peak; active peer-to-student ratio $\geq$ 5 per 100.
RO-02	Peer Counselor Misconduct	High	A peer provides harmful, inappropriate, or unprofessional interactions — potentially damaging student well-being.	Mandatory training module with scenario-based assessment before account activation; discreet 'Report Concern' button in every chat; automated account suspension after 3 distinct reports within 7 days.	Immediate suspension; reassign affected student to professional counselor; incident review within 24h.	$< 0.5\%$ of total sessions flagged for misconduct per semester.

RO-03	Peer Counselor Burnout	High	Overloaded peers experience emotional exhaustion, reducing interaction quality and increasing dropout.	Enforce maximum 3 sessions per day per peer counselor; automated wellness check-in prompts; peer wellness resources in the library.	Redistribute session load; recruit backup peers; offer formal wellness break without penalty.	Peer retention rate $\geq 70\%$ at 6-month mark; session cap compliance $\geq 95\%$.
RO-04	Low Student Adoption	High	Platform fails to reach minimum adoption (15% registration) in semester 1 due to low awareness or persistent stigma.	Multi-channel awareness campaign (social media, faculty announcements, orientation events, campus QR codes); anonymity-first messaging in all communications.	Reactivate campaign budget; activate student government partnership; offer facilitated first-session workshops.	$\geq 15\%$ of enrolled students registered by end of semester 1; $\geq 30\%$ by end of year 1.
SECURITY						
RS-01	User De-anonymization	High	A system flaw or malicious action links an anonymous platform profile to a real student identity.	Strict data segregation between SSO authentication tokens and anonymous profile IDs (zero-knowledge architecture); Privacy Impact Assessment (PIA) conducted before every release.	Immediate investigation; notify affected user; apply patch; document per Ley 1581 de 2012 data breach protocol.	Zero identity exposure incidents per semester.
RS-02	Account Takeover	High	Attacker gains control of a student or counselor account through credential stuffing or phishing.	MFA enforcement via SSO; anomalous login detection (IP and device fingerprinting); 30-minute inactivity session timeout; rate limiting on authentication endpoints.	Force-logout all active sessions; reset credentials; log and investigate incident.	Zero confirmed account takeovers per semester.

RS-03	Platform Exploitation / Predatory Use	High	Malicious users exploit the anonymous connection feature to manipulate or harm vulnerable students.	NLP-based keyword flagging for harmful content; strict peer verification process before activation; in-chat privacy reminders discouraging personal information sharing.	Immediate account ban; alert university security office; provide support to affected student.	Zero confirmed exploitation incidents per semester.
HUMAN AND SOCIAL						
RH-01	Harmful Peer Interaction / Emotional Damage	High	A peer counselor, despite training, gives advice that worsens a student's emotional state — whether through poor judgment, personal bias, or exceeding their role.	Scenario-based training with role boundaries clearly defined; peer accounts limited to emotional support (not clinical advice); supervisor review of flagged sessions within 24h.	Immediate session termination; student referred to professional counselor; mandatory debrief with supervisor.	Zero incidents where a peer interaction is assessed as clinically harmful; < 0.5% sessions flagged per semester.
RH-02	Exploitation of Vulnerable Users	High	A bad actor joins the platform as a peer counselor or student to manipulate emotionally vulnerable users — building dependency, extracting personal information, or initiating inappropriate relationships.	Identity verification before peer activation; NLP-based anomaly detection on conversation patterns; session-length and frequency limits to prevent dependency formation; in-chat privacy reminders.	Immediate account suspension; notify university security office; provide psychological first aid to affected student.	Zero confirmed exploitation incidents per semester; all peer accounts verified before activation.
RH-03	Loss of Trust in the Platform	High	A rumor, incident, or perceived privacy failure — even if unverified — triggers mass user abandonment.	Proactive transparency: publish a plain-language privacy policy; communicate any incidents honestly and promptly;	Issue a public platform statement within 48h; conduct independent privacy audit; engage student government to	Monthly active user rate does not drop more than 10% following any reported incident; $\geq$ 80% of users

			Trust in a mental health platform is fragile and hard to rebuild once broken.	designate a student-facing trust contact for concerns.	restore confidence.	rate the platform as trustworthy in semester surveys.
PROJECT MANAGEMENT						
RP-01	Development Timeline Overrun	Medium	Core modules not delivered on schedule due to complexity underestimation or resource gaps.	Two-week Agile sprints with 15% built-in time buffer; bi-weekly scope reviews; PMBOK-aligned change control process.	Descscope lowest-priority features for MVP release; allocate additional developer resources.	All milestones delivered within ± 1 sprint of the baseline schedule.
RP-02	Stakeholder Disengagement	Low	University administration or counseling department reduces cooperation during implementation.	Monthly steering committee meetings; defined RACI matrix; formal MoU with counseling department from project initiation.	Escalate to university IT committee; renegotiate integration scope and timelines.	All formal stakeholder sign-offs completed on schedule per project plan.

Risk Management Matrix and Mitigation Strategy Visualization

The heat map below places all 13 identified risks on the probability-impact grid. Color indicates composite score. Risk IDs link to the full descriptions in Section “Risk Descriptions, Priorities, and Mitigation Actions”.

Prob. \ Impact	Low	Medium	High	Critical
High	—	—	RO-01 · RO-03 · RO-04	—
Medium	—	RP-01	RT-01 · RT-04 · RO-04 · RS-02 · RH-03	RO-02 · RH-01
Low	RP-02	—	—	RT-02 · RT-03 · RS-01 · RS-03 · RH-02

CRITICAL — 1 risk RO-01	HIGH — 12 risks RT-01–04 · RO-02,03,04 · RS-01,02 · RH-01,02,03	MEDIUM — 2 risks RP-01 · RS-03 (by module)	LOW — 1 risk RP-02
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The table below maps each composite score to the system module it most directly affects, showing where mitigation investment is concentrated.

System Module	Risk IDs	Max Score	Mitigation Focus
Anonymous Communication Module	RT-03 · RO-02 · RS-03 · RH-01 · RH-02	High	E2E encryption; peer behavior governance; NLP content monitoring; exploitation detection.
Matching Engine / Peer Assignment	RO-01 · RO-03	Critical	Predictive recruitment; session caps; 20% demand buffer pool.
User Registration Module	RT-01 · RO-04 · RS-02	High	SSO resilience; token caching; login anomaly detection; onboarding campaign.
Privacy & Consent Module	RS-01	High	Zero-knowledge architecture; strict token-ID separation; PIA per release.
Infrastructure Layer	RT-02 · RT-04	High	Multi-AZ failover; daily backups; replication; auto-scaling.
Trust / Platform Reputation	RH-03	High	Proactive transparency; incident communication protocol; student-facing trust contact.
Project Management	RP-01 · RP-02	Medium	Agile sprints; RACI accountability; formal MoU; monthly steering committee.

Evolution Summary

The platform did not emerge fully formed. Each workshop produced something concrete that the next one had to build on or correct. This section documents that progression honestly — what each phase got right, what it left incomplete, and how the next phase closed the gap.

Workshop 1 → Workshop 2: Survey data becomes architecture

Workshop 1 finding	Workshop 2 decision	Reasoning
88% of students reported anxiety; 68% had never sought help.	Tiered support model: self-help → peer chat → professional counseling.	Students needed a low-friction entry point. The tiered model avoids overloading professional counselors while still offering escalation for serious cases.
64% preferred anonymous support. Zero students rated existing services as easy to access.	Privacy-first architecture: anonymized SSO profiles; zero-knowledge chat layer; Privacy & Consent module as a cross-cutting component.	Anonymity is the primary trust enabler. Without it, the platform would replicate the same barriers that currently stop students from seeking help.
Demand spikes are predictable around midterms and finals.	Auto-scaling cloud infrastructure + predictive peer recruitment triggered 3 weeks before peak periods.	Infrastructure and human capacity need to scale together. One without the other creates a gap precisely when the platform is needed most.

One harmful peer interaction can cascade into mass abandonment (nonlinear effect identified in the W1 complexity analysis).	Mandatory training module; in-chat Report button; 3-report auto-suspension of peer accounts.	The risk is nonlinear: a single bad interaction spreads through word of mouth. Peer governance had to become a baseline requirement, not an optional feature.
68% wanted all types of support combined — professional, peer, and educational resources.	Five integrated modules: Matching Engine, Anonymous Communication, Appointment Scheduling, Resource Library, Privacy & Consent.	Students want one platform for everything. Siloed services reproduce the same navigation barriers that currently prevent help-seeking behavior.

Workshop 2 → Workshop 3: Architecture becomes production-ready

What Workshop 2 had	What Workshop 3 added	Why it changed
General cloud hosting as the infrastructure strategy.	Multi-AZ failover with $RPO \leq 1h$ and $RTO \leq 2h$ formally committed.	Good architectural intent without measurable SLA targets is not enough for a system that must be available precisely when students are most stressed.
Sensitivity factors identified with descriptive mitigations — no thresholds.	ISO 31000-aligned risk register with 13 risks and a measurable acceptance criterion per mitigation (Section 3).	A mitigation without a success criterion is not a control — it is a wish. Every risk now has a number attached to it and an owner accountable for it.
Modular architecture with REST APIs. Resilience mentioned but not specified.	Fault-tolerance per module: circuit breaker patterns; queue-based messaging for chat; graceful degradation so one module failure does not cascade.	ISO 25010 reliability attributes were absent from W2. If the chat module fails, appointments and the resource library should keep running independently.
Continuous improvement cycle outlined in W2 §6.3: feedback loop, data-driven adjustments, adaptive capacity planning.	Each improvement trigger is now tied to a specific metric threshold and a responsible owner (see Section 3 acceptance criteria).	W2 described the cycle correctly but left it abstract. W3 closes the loop: when a metric falls below threshold, the response — and who executes it — is already defined.
Privacy described as a design principle.	Privacy Impact Assessment (PIA) mandated before every release; Ley 1581 de 2012 compliance documented as a formal acceptance criterion (RS-01).	Regulatory compliance requires documented evidence, not architectural intent. A PIA report is the proof that the zero-knowledge design is actually working.

Key Lessons Learned

Source	Lesson	Effect on the design
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Survey data	Lack of awareness is a bigger barrier than stigma. 56% of students did not know mental health services existed.	Platform visibility became a primary requirement. Registration rate ($\geq 15\%$ in semester 1) is now a formal KPI with a defined measurement method, not an aspiration.
Complexity analysis	The system is more fragile than it looks. One bad peer interaction can undo months of built trust.	Peer governance was elevated from a nice-to-have to a non-negotiable baseline control with its own acceptance criterion ($< 0.5\%$ of sessions flagged per semester).
Architecture design	Strong anonymity and effective moderation are in tension.	The zero-knowledge architecture — where SSO identity and platform profile are separated at the database level — resolves the tension without sacrificing either property.
Continuous improvement	A feedback loop without thresholds cannot drive improvement. It just generates data.	Every monitoring metric from W2 now has an explicit acceptance criterion in W3. When the metric drops, the response — and the owner — are already defined.
Risk analysis	Low-probability / critical-impact risks need the same investment as frequent ones.	RT-03 (Encryption Breach) and RS-01 (De-anonymization) received quarterly pen-testing and mandatory PIA despite their low probability, because failure is irreversible.

Design Evolution

Dimension	Workshop 1	Workshop 2	Workshop 3
System scope	Problem diagnosis; survey; stakeholder mapping.	Three-layer architecture; 6 modules; REST APIs.	Fault-tolerant modules; measurable SLA targets; governance structure.
Risk handling	Sensitivity factors identified qualitatively.	Descriptive mitigations per sensitivity factor.	ISO 31000 register; 13 risks; acceptance criteria per mitigation.
Continuous impr.	Not addressed.	Feedback loop, data-driven adjustments, and adaptive capacity outlined (§6.3).	Each trigger tied to a metric threshold and a responsible owner.

Privacy	Anonymity identified as critical requirement.	Zero-knowledge design; E2E encryption mandated.	PIA before every release; Ley 1581 compliance as formal criterion.
Quality assurance	Performance observations from survey data.	Metrics and monitoring approach defined.	Acceptance criteria per risk; QA framework completed by Person 3.
Project management	Not addressed.	Informal phased roadmap.	Project Charter; RACI; Gantt; risk-adjusted schedule (Person 4).

Quality Assurance Framework and Project Management

This section presents the Quality Assurance (QA) framework, validation processes, Project Charter, and team structure for the Student Mental Health Support Platform. The content builds directly on the system analysis and architectural design, and aligns with the robust design principles, quality standards, and project management methodologies.

Quality Assurance Framework

The Quality Assurance Framework defines the systematic activities, metrics, and standards that will ensure the platform meets its quality objectives and stakeholder expectations. Given the sensitive nature of mental health support, the framework particularly emphasises privacy, reliability, usability, and maintainability.

Quality Standards and Guidelines Adopted

- ISO 9001 (Quality Management Systems):
Provides the overarching process-based approach for continual improvement, customer focus, and evidence-based decision-making. All design and development processes will be documented, reviewed, and improved iteratively.
- ISO/IEC 27001 (Information Security Management):
Fundamental to the platform's requirement for anonymity and data protection. This standard governs risk assessment, security controls, and incident response for the encrypted databases, end-to-end encrypted chat, and authentication interfaces.
- CMMI (Capability Maturity Model Integration):
Used to assess and improve the maturity of the development process, particularly in requirements management, verification, and validation. The goal is to operate at a managed process level where quality is planned, monitored, and controlled.
- Six Sigma Principles:
Applied to reduce defects in critical processes, such as the matching engine accuracy and appointment scheduling reliability. Statistical analysis of user feedback and system logs will drive root-cause corrections.

The adoption of these standards ensures that the quality framework is both structured and aligned with industry best practices.

Quality Metrics and Performance Indicators

Quantifiable metrics have been established to objectively measure system quality. These metrics derive directly from the non-functional requirements identified and are aligned with the performance metrics defined in the system design document.

Quality Attribute	Metric	Target / Success Criterion	Measurement Method
Performance	Page load time (P95)	≤ 2 seconds under normal load	Automated browser testing (Lighthouse, JMeter)
	Time to initial peer match (P90)	≤ 10 minutes during peak hours	Timestamp logs in Matching Engine
Reliability	System uptime during active semesters	$\geq 99\%$	Infrastructure monitoring (AWS CloudWatch or equivalent)
	Graceful degradation – core modules remain available if non-critical module fails	100% partial functionality maintained	Fault injection testing; module isolation design
Scalability	Handle 300% increase in concurrent users without exceeding response time thresholds	P95 page load $\leq 3s$, match time ≤ 15 min during simulated surge	Load testing with auto-scaling validation
Usability	Task success rate for first-time registration	$\geq 90\%$	Usability testing with 15–20 student volunteers
	User satisfaction rating (post-session)	$\geq 4.0 / 5.0$	In-app survey aggregation
Security & Privacy	Successful penetration test findings (critical/high severity)	0 critical or high vulnerabilities prior to go-live	Independent security audit and automated scanning (OWASP ZAP)
	Data encryption compliance (end-to-end chat, database at rest/in transit)	100% compliant with ISO 27001 controls	Configuration audits; certificate verification
Maintainability	Module update deployment without system downtime	100% independent module updates with zero downtime	Rolling update testing in staging environment

These metrics will be continuously tracked through a live monitoring dashboard and reviewed during each project phase.

Acceptance Criteria

To confirm that the system is ready for pilot deployment, the following module-level acceptance criteria must be satisfied:

Module	Acceptance Criteria
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User Registration & Authentication	University SSO integration works; anonymous profiles created without storing identifiable data beyond the authentication token; duplicate or invalid registration attempts handled gracefully.
Matching Engine	At least 90% of students completing the needs assessment receive a peer match within 10 minutes under baseline load; matching quality feedback averages $\geq 3.5/5$ after the first two weeks of pilot.
Anonymous Communication	Chat messages are encrypted end-to-end; messages are not logged in plaintext; real-time delivery under normal network conditions with latency < 500 ms.
Appointment Scheduling	Students can select available time slots and receive confirmation within 48 hours; no double-booking occurs; professional counselors can view and manage their schedules.
Resource Library	Administrators can upload, categorise, and remove content; all users can search and view resources with page load < 2 seconds.
Privacy & Consent Management	Privacy reminders appear in chat; users are prompted for consent on first use; reporting and moderation workflows function correctly.

Validation and Verification Methods

The QA framework employs both verification and validation across the development lifecycle:

- Reviews and Inspections:
 - Design documents, code, and configuration are peer-reviewed against the system requirements and the ISO 9001 quality plan. Architectural gate checks are held at the end of each major design refinement.
- Testing Levels:
 - Unit Testing: All backend modules (Python) and frontend components (React) are covered by automated unit tests, with code coverage goals ($\geq 80\%$).
 - Integration Testing: REST API contracts and inter-module communication are verified using mocked external services (university SSO, notification service).
 - System Testing: End-to-end scenarios (registration to session feedback) are executed in a staging environment that mirrors production.
 - Performance and Stress Testing: Simulated loads up to 300% of expected peak are applied using tools like Locust or JMeter; auto-scaling behaviour is observed.
 - Security Testing: Static code analysis, dependency checks, and external vulnerability scans are performed before every release. An external penetration test is scheduled before the pilot launch.
 - User Acceptance Testing (UAT): A selected group of students, peer counsellors, and administrators will perform real tasks during the pilot phase; their feedback directly validates usability and satisfaction metrics.
- Validation through Continuous Feedback:
 - Post-session ratings, monthly satisfaction surveys, and usage analytics are compared against the defined targets. Deviations trigger corrective actions in the continuous improvement cycle.

Monitoring and Control Mechanisms

Quality assurance continues into operations through the following mechanisms:

- **Real-time Monitoring Dashboard:**
Displays key metrics (response times, match queue length, server health, error rates) with automated alerts when thresholds are breached.
- **Automated Error Reporting:**
Exceptions and failed requests are logged with traces; a severity-based escalation protocol notifies the platform administrators.
- **Periodic Quality Audits:**
Every semester, a review of logs, incident reports, and user feedback is conducted against ISO 9001 and ISO 27001 checklists; improvements are planned and tracked through the project's issue tracker.

Quality Assurance Process Flows and Validation Frameworks

This subsection describes the end-to-end process flow that integrates quality assurance activities from design to continuous operation. The diagram below illustrates the full QA lifecycle for the Student Mental Health Support Platform.

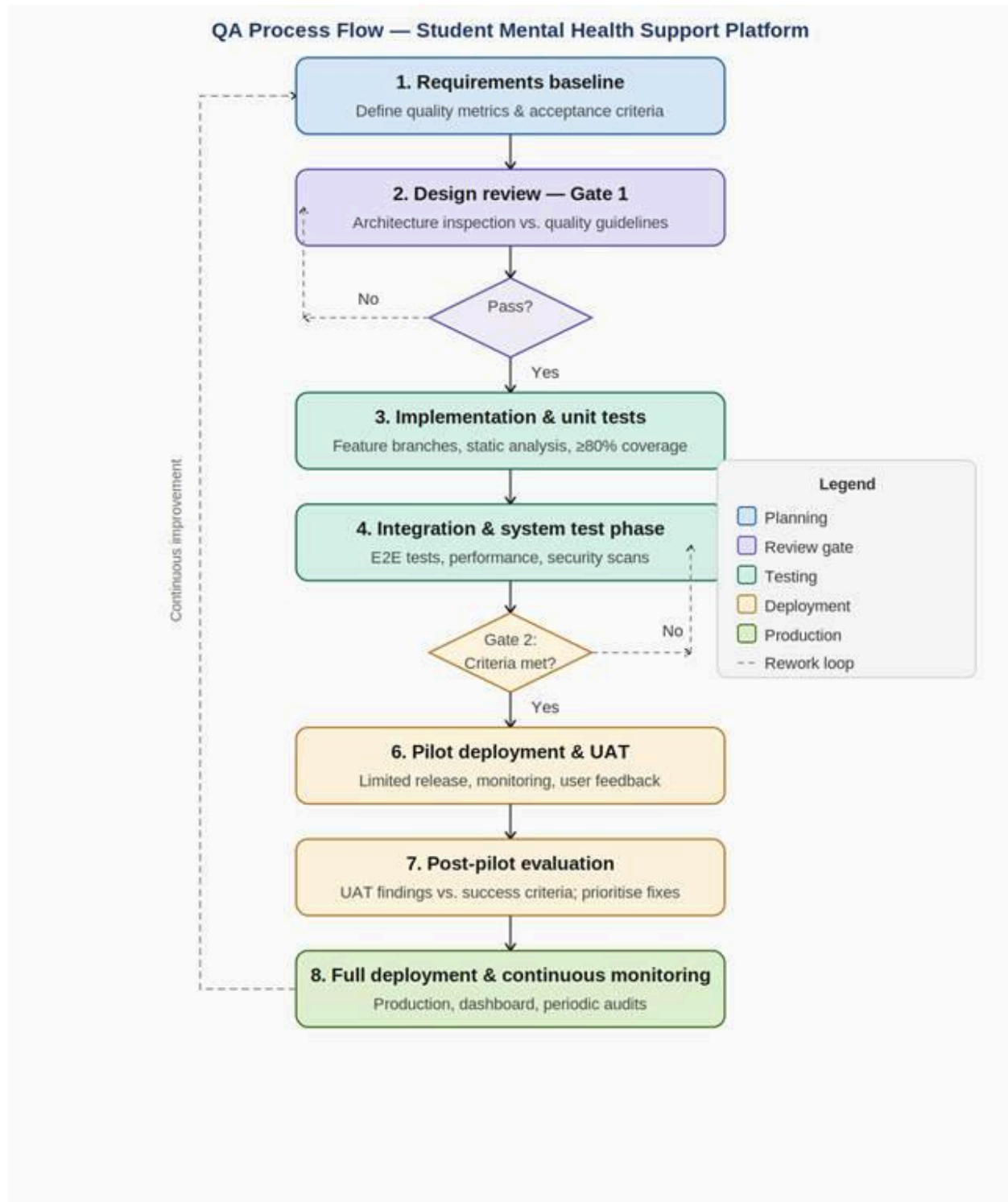


Figure 1. Quality Assurance Process Flow — Student Mental Health Support Platform

The QA process flow comprises the following sequential phases:

- Requirements Baseline: Functional and non-functional requirements are reviewed; quality metrics and acceptance criteria are defined (as in Section 5.1.2).
- Design Review Gate 1: Architecture and module designs are inspected for compliance with quality guidelines (modularity, fault-tolerance, scalability) and security/privacy controls.
- Implementation & Unit Tests: Code is developed in feature branches; automated unit tests and static analysis gate pull requests.
- Integration & System Test Phase: Backend and frontend are integrated; automated integration tests, performance tests, and security scans are run. Results recorded in a test management tool.
- Pre-Pilot Validation Gate 2: Acceptance criteria (Section 5.1.3) must be met in the staging environment. A go/no-go decision is made by the project manager and QA lead.
- Pilot Deployment & UAT: Limited user release with monitoring and feedback collection.
- Post-Pilot Evaluation: UAT findings, performance data, and user satisfaction metrics are analysed against success criteria. Adjustments are prioritised for the next iteration.
- Full Deployment & Continuous Monitoring: The system enters production under the monitoring controls described in Section 5.1.5. The monitoring dashboard and periodic audits close the loop, feeding improvements back to the requirements baseline.

Validation Framework per Quality Attribute

The following table summarises the validation techniques, frequency, and responsibilities for each quality attribute:

Quality Attribute	Validation Technique	Frequency	Responsible Role
Performance	Load testing, synthetic monitoring	Every release; continuous monitoring	QA Lead
Reliability	Fault injection, uptime monitoring	Pre-pilot and after major changes	System Architect
Scalability	Auto-scaling tests with production-scale simulations	Before pilot and before each semester start	System Architect
Usability	Task-based user testing, SUS questionnaire, post-session surveys	Pilot phase and annually	QA Lead
Security	Automated scanning, manual penetration testing, configuration audits	Every release; external pen test before go-live	QA Lead
Maintainability	Deployment drills, module isolation testing	Every release	System Architect

These validation activities are scheduled in the project timeline and tracked via the quality register.

Project Charter

The Project Charter formally authorises the Student Mental Health Support Platform – Robust Implementation Planning project and provides a high-level overview of scope, objectives, stakeholders, and success criteria. It serves as the reference point for all subsequent planning and execution.

Project Title: Student Mental Health Support Platform

Project Purpose: To refine the system design developed in Workshop 2 with robust engineering principles, integrate comprehensive quality assurance and risk management frameworks, and produce a detailed project management plan that prepares the platform for pilot implementation.

In Scope:

- Enhancement of the system architecture to incorporate fault-tolerance, scalability, and maintainability.
- Development of a Quality Assurance Framework (metrics, acceptance criteria, validation flows).
- Creation of a Risk Management Plan (technical, operational, security, project risks) with mitigation strategies.
- Definition of a Project Management Plan including team roles, schedule, resources, and implementation strategy.

Out of Scope:

- Actual coding or deployment of the platform (these are future phases).
- Detailed business case or procurement activities beyond resource estimation.
- Clinical validation or medical device compliance.

Project Objectives

- Deliver a robust design that guarantees 99% availability, sub-2-second response times, and the ability to handle a 300% demand surge without degradation.
- Achieve full traceability from Workshop 1 findings to quality metrics and risk mitigations.
- Establish a realistic and phased implementation roadmap including pilot and full-scale rollout.
- Obtain formal approval (from the course professor) that the design and management plans are feasible and academically sound.

Key Stakeholders

- Primary Users: University students, peer counsellors, professional counsellors.
- University Administration: Service quality oversight, resource library content provision, policy compliance.
- Platform Administrators: Technical operation and maintenance of the final system.
- Course Evaluator: Eng. Carlos Andrés Sierra, M.Sc., as the instructor assessing the deliverables.

Success Criteria

- All quality metrics defined in Section are integrated into the design with verifiable validation plans.
- A complete risk register (following PMBOK/ISO 31000) is delivered, containing at least 15 identified risks with detailed mitigation strategies.
- The project management plan includes a Gantt chart with realistic milestones, a clear team allocation, and a communication plan.
- The enhanced system architecture and QA/risk frameworks are accepted in the final submission evaluation.

High-Level Milestones

Timeline	Milestone	Deliverable / Outcome
Weeks 1–2	Architecture Refinement	Updated architecture incorporating fault-tolerance, scalability, and maintainability principles and standards alignment.
Weeks 3–4	QA & Risk Plan Development	Complete Quality Assurance Framework and Risk Management Plan, including visual diagrams.
Week 5	Project Management Plan Assembly	Integrated project management plan with Gantt chart, resource plan, and communication plan; internal review completed.
Week 6	Final Integration & Submission	All documents integrated into a single PDF deliverable; GitHub repository updated; Workshop 3 submission completed.

Constraints and Assumptions

- The platform will be hosted on a university-approved cloud provider; the engineering is based on Python/React/PostgreSQL.
- The team of four students has equal workload distribution; external expert review is not available.
- The academic calendar of 2026-I provides the timeline boundaries.

Approval: This charter is endorsed by the project team and is subject to acceptance by the course professor as part of the Workshop 3 evaluation.

Team Structure and Roles

The project is carried out by the same four-member group. For the robust design and project management phase, roles have been formally assigned to ensure clear accountability and alignment.

Organisational Chart

- Project Sponsor / Evaluator: Eng. Carlos Andrés Sierra, M.Sc. (Professor)
- Project Team (Execution Level):
 - Project Manager: Alan Santiago Agudelo Sarmiento
 - Quality Assurance & Risk Manager: Juan David Cardozo Trujillo
 - System Architect & Technical Lead: Julian Ernesto Romero Gutierrez
 - Documentation & Integration Lead: Nicolas Alexander Sierra Contreras

All team members collaborate on design and review activities, but the designated leads are responsible for the completion and integrity of their respective domains.

Role Descriptions and Responsibilities

Role	Assigned To	Key Responsibilities	Accountability
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Project Manager	Alan S. Agudelo S.	Overall project coordination; project charter; schedule (Gantt chart); stakeholder communication; integration of all sections into the final deliverable.	Project delivered on time and meeting Workshop 3 requirements.
QA & Risk Manager	Juan D. Cardozo T.	Development of the Quality Assurance Framework (metrics, acceptance criteria, validation flows); risk identification and mitigation planning; QA process diagrams.	QA and risk sections are complete, consistent, and aligned with ISO 9001/27001
System Architect & Technical Lead	Julian E. Romero G.	Enhancement of system architecture diagrams to reflect robustness principles; technical feasibility analysis; alignment with quality guidelines (ISO, CMMI); technical review of risk mitigations.	Architecture design is technically sound and robust, incorporating modularity, fault-tolerance, and scalability.
Documentation & Integration Lead	Nicolas A. Sierra C.	Repository management (GitHub structure, README updates); integration of cross-module content; professional formatting of the final PDF; version history and change log.	All documents are coherent, properly formatted, and the repository is up-to-date

Project Timeline

The project timeline covers the Student Mental Health Support Platform, beginning with system analysis and ending with production monitoring. The sequence is designed around a progressive lifecycle: discovery, requirements, design, robustness planning, MVP construction, validation, pilot deployment, and full rollout. The critical path runs through requirements baseline, architecture baseline, MVP core module completion, integration and security validation, pilot approval, and full rollout readiness.

Phase	Relative Weeks	Main Objective	Key Deliverables	Dependencies / Critical Path
1. Analysis and Discovery	W1-W3	Understand the student mental health problem, map stakeholders, collect survey data, and define system boundaries.	Survey results; stakeholder map; in-scope/out-of-scope definition; initial sensitivity and complexity analysis.	Project start. Critical because all later requirements depend on validated student needs.
2. Requirements Baseline	W3-W4	Translate Workshop 1 findings into functional, non-functional, privacy, and stakeholder requirements.	Requirements matrix; priority list; acceptance assumptions for anonymity, accessibility, performance, and peer availability.	Depends on survey interpretation and scope approval.

3. Architecture and System Design	W4-W6	Define the modular architecture and module responsibilities for the platform.	Three-layer architecture; module responsibility table; internal and external integration points; technology stack recommendation.	Depends on requirements baseline. Critical path item.
4. Robust Design, QA, and Risk Planning	W7-W9	Strengthen the design with reliability, scalability, maintainability, privacy, risk, and quality controls.	Risk register; quality assurance framework; validation gates; measurable metrics; project charter and team role allocation.	Depends on architecture baseline. Establishes implementation governance.
5. Repository Integration and Readiness Review	W9-W10	Organize all project documentation and prepare the repository for development and evaluation.	Updated README; folders for all workshops; architecture diagrams; risk register; QA checklist; project management exports.	Depends on completion of Workshops 1-3 documentation.
6. Infrastructure and Security Baseline	W11-W12	Prepare the technical environment for the MVP while enforcing security and privacy constraints.	Cloud environment; PostgreSQL database; SSO integration plan; encrypted storage; CI/CD skeleton; environment variables policy.	Depends on approved design and repository structure.
7. MVP Core Module Development	W13-W16	Build the minimum viable platform modules needed for a realistic pilot.	Registration; needs assessment; matching engine; anonymous chat; appointment scheduling; resource library; admin basics.	Critical path. Depends on infrastructure setup and module interface definitions.
8. Integration, QA, and Security Validation	W17-W18	Validate the MVP against quality criteria before exposing it to pilot users.	Unit/integration tests; load test for peak demand; OWASP ASVS-based security checklist; privacy impact review; go/no-go report.	Depends on MVP completion. Critical path validation gate.
9. Pilot Deployment and UAT	W19-W20	Release the platform to a limited group of students, peer counselors, and professional counselors.	Pilot launch; UAT feedback; post-session survey results; system performance dashboard; issue log.	Depends on validation gate approval.
10. Feedback-Driven Refinement	W21-W22	Correct usability, matching, performance, and communication issues detected during the pilot.	Updated backlog; resolved defects; adjusted matching rules; updated training and resource material.	Depends on pilot data and stakeholder review.
11. Full Rollout and Continuous Monitoring	W23-W24+	Expand use progressively across the university and operate the platform through monitoring and improvement cycles.	Release plan; adoption campaign; production dashboard; monitoring cadence; continuous improvement register.	Depends on pilot success and readiness approval.

Milestone Control Points

Milestone	Target Point	Acceptance Evidence	Primary Owner
M1 - Problem and stakeholder baseline approved	End W3	Survey data summarized; stakeholders and system scope documented.	Project Manager
M2 - Architecture baseline approved	End W6	Layered architecture, modules, interfaces, and technology stack documented.	System Architect & Technical Lead
M3 - Risk and QA baseline approved	End W9	Risk register, acceptance criteria, validation framework, and quality metrics completed.	QA & Risk Manager
M4 - Repository readiness achieved	End W10	All workshop documentation and project management materials available in the GitHub repository.	Documentation & Integration Lead
M5 - MVP feature complete	End W16	Core modules implemented in the development/staging environment.	System Architect & Technical Lead
M6 - Pre-pilot validation gate passed	End W18	Functional, performance, security, privacy, and usability checks passed or formally accepted with mitigations.	QA & Risk Manager
M7 - Pilot completed	End W20	Pilot feedback collected; performance and adoption data reviewed.	Project Manager
M8 - Full rollout approved	End W24	Defects closed, resource capacity confirmed, support process active, and monitoring dashboard operational.	University Admin / Project Manager

Project Timeline Visualization

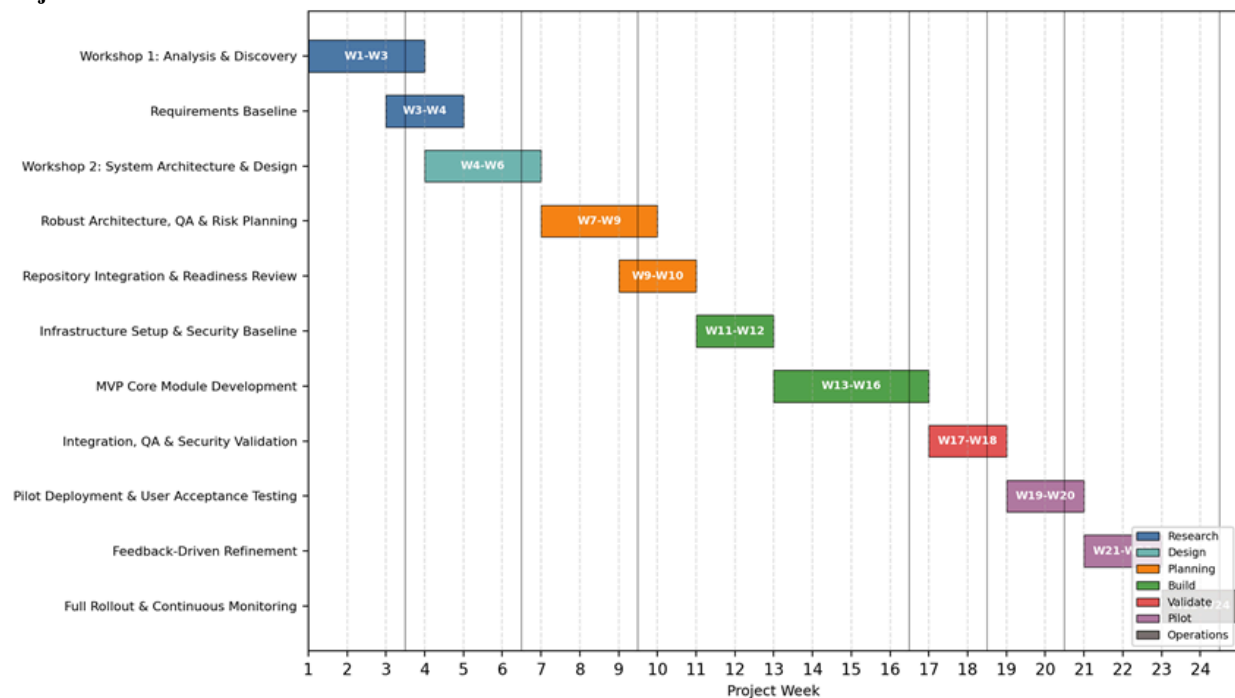


Figure 1. Project Gantt chart

Implementation Strategy

The implementation strategy follows a phased and risk-controlled approach. The system should first be delivered as a secure MVP focused on the most critical user journey: authenticated anonymous access, needs assessment, peer matching, anonymous communication, professional appointment scheduling, and resource access. The MVP is then validated through controlled pilot deployment before full rollout.

Implementation Phase	Description	Main Activities	Exit Criteria
Phase 1 - Requirements Freeze and Backlog	Convert workshop findings into a prioritized development backlog for the whole system.	Confirm functional requirements; classify MVP vs. later enhancements; define acceptance criteria for privacy, performance, peer availability, and usability.	Backlog approved; requirements traceable to Workshops 1 and 2; no critical ambiguity in core modules.
Phase 2 - Technical Foundation	Prepare the secure infrastructure needed to host and test the platform.	Configure cloud environment; create PostgreSQL schema; define SSO integration; establish CI/CD; create development, staging, and production branches/environments.	Environment available; database and access control validated; security baseline documented.
Phase 3 - MVP Development	Develop the platform modules needed for pilot use.	Build registration, anonymous profile generation, assessment, matching, chat, scheduling, resource library, admin dashboard, and privacy/consent controls.	Core user journey works end-to-end in staging.
Phase 4 - Integration and Validation	Validate that modules work together and satisfy quality objectives.	Run functional, integration, load, usability, and security tests; review OWASP ASVS controls; test graceful degradation and backup recovery.	Pre-pilot go/no-go checklist approved.
Phase 5 - Pilot Deployment	Deploy to a limited group to validate real-world behavior with controlled risk.	Onboard pilot users and peer counselors; monitor response time, wait time, feedback, adoption, and flagged interactions.	Pilot success criteria met or remediation plan approved.
Phase 6 - Refinement and Full Rollout	Improve the system based on pilot evidence and expand by faculty or student group.	Fix defects; tune matching; increase peer counselor capacity; launch communication campaign; activate support procedures.	Full rollout approved; monitoring and incident response active.
Phase 7 - Continuous Operation	Operate the system with measurable quality, risk, and improvement controls.	Review dashboards; analyze feedback; update resources; schedule security reviews; plan feature increments such as analytics or NLP-assisted matching.	Metrics remain within thresholds and improvement backlog is updated monthly.

Deployment and Rollout Strategy

- Start with a staging environment that mirrors production but contains test data only, protecting privacy while enabling realistic validation.
- Use a limited pilot before institutional rollout because the system involves sensitive mental health interactions and peer counselor behavior cannot be fully predicted in advance.
- Scale both technical infrastructure and human support capacity. Auto-scaling alone is insufficient if peer counselor availability does not increase during midterms and finals.
- Apply modular releases. Registration, matching, chat, scheduling, and resource library modules should be deployed independently when possible, reducing downtime and isolating failures.

Include training and adoption as implementation work, not as a final communication task. Peer counselor training, privacy messaging, and student awareness campaigns directly determine system success.

Technical and Operational Prerequisites

Prerequisite	Reason	Required Before
University SSO or approved authentication method	Students must be verified as university members while their public platform identity remains anonymous.	MVP pilot
Encrypted database and separated identity/profile storage	Protects anonymity and reduces de-anonymization risk.	MVP development
Trained peer counselor pool	The platform depends on safe, available, and supervised human support.	Pilot deployment
Counseling department schedule availability	Professional appointment scheduling requires updated availability windows and escalation paths.	Pilot deployment
QA and security checklists	Prevents sensitive functions from being released without validation evidence.	Pre-pilot gate
Monitoring dashboard	Supports operational control over uptime, response time, adoption, wait time, and flagged sessions.	Pilot and full rollout

Resource Management Plan

Role / Resource	Assigned / Responsible Area	Whole-Project Responsibilities	Estimated Allocation	Key Outputs
Project Manager	Alan Santiago Agudelo Sarmiento	Coordinate the entire project lifecycle; maintain schedule; manage stakeholder communication; integrate analysis, design, QA, and implementation plans; prepare status reports.	Medium-High during planning and gates; Medium during development.	Project charter, timeline, communication plan, status reports, rollout readiness decisions.
QA & Risk Manager	Juan David Cardozo Trujillo	Maintain the risk register; define acceptance criteria; coordinate testing strategy; monitor quality thresholds; manage issue escalation.	High during QA/risk and validation phases; Medium during operation.	Risk register, QA checklist, test evidence, go/no-go reports.
System Architect & Technical Lead	Julian Ernesto Romero Gutierrez	Own architecture decisions; define module interfaces; guide backend/frontend integration; validate scalability, maintainability, and reliability.	High during design and MVP development.	Architecture diagrams, interface specifications, deployment plan, technical review notes.
Documentation & Integration Lead	Nicolas Alexander Sierra Contreras	Maintain repository structure; update README; version documents; ensure traceability across workshops; organize diagrams, exports, and supporting material.	Medium across all phases; High before submissions/releases.	Repository update, README, change log, documentation index, exported PDFs.

Peer Counselors	Operational resource	Provide safe peer support through anonymous communication; comply with training, session limits, and reporting protocols.	Variable; increased during exam periods.	Available peer support sessions; feedback data; adherence to session caps.
Professional Counselors	Counseling service resource	Handle escalated support needs and scheduled appointments; participate in pilot feedback and escalation protocols.	Part-time, scheduled availability.	Appointment slots, escalation response, professional feedback.
University Administration	Governance resource	Approve policies; provide content for resource libraries; support adoption campaigns; ensure compliance with institutional rules.	Low-Medium, higher at gates.	Policy approvals, content validation, campaign support.
Platform Administrators	Operational/technical resource	Monitor uptime, logs, access control, backups, and incident response.	Medium during pilot and production.	Monitoring reports, incident logs, maintenance records.

Capacity and Budget Considerations

Resource Category	Purpose	Capacity / Cost Assumption	Control Mechanism
Cloud hosting and database	Host backend, frontend, encrypted database, backups, monitoring, and staging environment.	Start with low-cost academic/cloud tier; scale before exam periods; budget should reserve capacity for 300% demand surge testing.	Monthly usage review; auto-scaling thresholds; RPO/RTO drills.
Development tools	Support coding, version control, CI/CD, diagrams, tests, and documentation.	Use GitHub, open-source frameworks, and university-approved tools where possible.	Repository audit; branch protection; pull request templates.
Security and privacy validation	Protect anonymous communication and sensitive user information.	Internal checklist every release; external or instructor-reviewed security assessment before go-live if available.	OWASP ASVS checklist; privacy impact assessment; vulnerability tracking.
Peer counselor program	Ensure enough trained support providers for the platform to function.	Recruit before midterms/finals; maintain a 20% buffer above projected demand when possible.	Peer availability dashboard; session cap monitoring; retention tracking.
Training and documentation	Prepare students, peer counselors, professionals, and admins to use the system safely.	Create reusable onboarding guides, privacy resources, and role-based instructions.	Training completion records; user feedback; resource library review.
Communication campaign	Increase student awareness and adoption.	Use university channels, social media, orientation events, QR codes, and student organizations.	Registration rate; active users; campaign metrics.

RACI Summary for Main Deliverables

Deliverable / Activity	Responsible	Accountable	Consulted	Informed
Requirements baseline	Project Manager; Documentation Lead	Project Manager	Students, University Admin, Professional Counselors	Professor / Evaluator
Architecture and module interfaces	System Architect	System Architect	QA & Risk Manager, Platform Admins	Project Team
Risk register and QA framework	QA & Risk Manager	QA & Risk Manager	System Architect, University Admin	Project Team, Professor
MVP development	System Architect; Development contributors	Project Manager	QA & Risk Manager, Documentation Lead	Stakeholders
Pilot deployment	Project Manager; Platform Admins	Project Manager	University Admin, Peer Counselors, Professional Counselors	Pilot users
Repository and documentation	Documentation & Integration Lead	Project Manager	All team members	Professor / Evaluator

Communication and Control Plan

The communication and control plan ensures that all project stakeholders understand progress, risks, decisions, and responsibilities. Because the platform handles sensitive mental health interactions, communication must also protect confidentiality and avoid exposing personal or clinical details. Project controls are based on measurable quality indicators, risk review cadence, and formal decision gates.

Stakeholder / Audience	Information Needed	Channel / Artifact	Frequency	Owner
Project team	Task progress, blockers, design decisions, risk changes, repository updates.	Weekly meeting; GitHub issues; project board; shared documentation.	Weekly during planning/build; twice weekly during pilot.	Project Manager
Professor / evaluator	Evidence of continuity from Workshop 1 to 3, deliverables, project plan, repository link, and final documentation.	Formal document submission; repository README; milestone summary.	At academic submission points and major reviews.	Project Manager / Documentation Lead
University administration	Policy compliance, adoption progress, resource library status, incident summaries, and rollout readiness.	Steering review; dashboard summary; approval checklist.	Monthly during planning; bi-weekly during pilot.	Project Manager
Professional counselors	Appointment scheduling process, escalation rules, pilot feedback, and availability planning.	Role guide; meeting; scheduling dashboard.	Before pilot; weekly during pilot.	Project Manager / University Admin
Peer counselors	Training requirements, session limits, safe support rules, feedback expectations, and peak-demand notices.	Training module; in-app notifications; email reminders.	Before activation; three weeks before exam peaks; ongoing alerts.	QA & Risk Manager

Students / end users	Platform purpose, anonymity guarantees, how to access support, what is not provided, and how to report concerns.	Launch campaign; in-app onboarding; privacy resource hub; FAQ.	Before and during rollout; updated each release.	Documentation Lead / University Admin
Platform administrators	System health, incidents, logs, backups, deployment status, and access control changes.	Monitoring dashboard; incident report; deployment notes.	Continuous monitoring; weekly operations review.	System Architect

Project Control Mechanisms

Control Area	Metric / Evidence	Target or Threshold	Review Cadence	Escalation Rule
Schedule control	Milestone completion, sprint burndown, blocked tasks.	Milestones delivered within +/- one sprint of baseline.	Weekly.	If a critical-path task slips more than one week, project manager calls scope review.
Scope control	Approved backlog, change requests, traceability to requirements.	No new feature enters MVP without impact review.	Every sprint.	Unapproved scope changes move to post-pilot backlog.
Quality control	Test pass rate, defects by severity, validation checklist.	No critical defects open before pilot.	Every release.	Critical defects block go/no-go decision.
Performance control	Page load time, API latency, uptime, error rate.	Sub-2-second normal response target; 99% active-period uptime target.	Continuous; reviewed weekly.	Threshold violation triggers technical incident review.
Privacy and security control	OWASP ASVS checklist, vulnerability scan, privacy impact assessment, access review.	No unresolved high-severity vulnerability before release.	Every release; quarterly deeper review.	High severity issue blocks release until mitigated.
Operational capacity control	Active peer-to-student ratio, wait time, session cap compliance.	95% of peak users matched or routed within acceptable wait target; session caps $\geq$ 95% compliance.	Weekly; increased during exam peaks.	Shortage triggers recruitment campaign and professional counselor fallback.
User trust and adoption control	Registration rate, monthly active users, survey trust score, reported concerns.	Semester 1 registration target of at least 15%; trust score monitored after incidents.	Monthly.	Adoption/trust drop triggers communication campaign and privacy clarification.
Repository control	README completeness, folder structure, version history.	All workshops and management materials	Before each submission/release.	Repository defects must be corrected before final submission.

	deliverable formats.	organized and linked.		
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Decision Gates

- Gate 1 - Requirements and Architecture Approval: confirms the system still responds to Workshop 1 findings and Workshop 2 design constraints.
- Gate 2 - Pre-Pilot Validation: confirms functional, security, performance, privacy, and usability evidence is sufficient for limited user exposure.
- Gate 3 - Full Rollout Approval: confirms pilot defects are resolved, peer counselor capacity is adequate, and operational monitoring is active.
- Gate 4 - Continuous Improvement Review: evaluates monthly metrics, incident history, user feedback, and repository updates to prioritize the next improvement cycle.

Github Repository

The repository was updated to reflect the progress of the complete Student Mental Health Support Platform project. A new folder named Workshop_3_Management was created to organize all materials related to the robust system design and project management phase. This folder contains two subfolders: docs, where the final project management document is stored in PDF format, and diagrams, where all new PDF diagrams generated for this phase are included. This structure improves repository navigation and documentation clarity, aligning with GitHub's recommendation to keep repository documentation organized so users can understand and navigate the project more easily.

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